

1. Write a Python program to append a list to the second list.
2. Python program to swap two numbers using function
3. Write a Python program to create a Vehicle class with max_speed and mileage instance Attributes.
4. Write a python program to Create a child class Bus that will inherit all of the variables and methods of the Vehicle class
5. Print the given number in words.(eg.1234 => one two three four). numbers = ["", "one", "two", "three", "four"]
6. Python program to find sum of n numbers using for loop
7. Using for loop, write and run a Python program to find factorial from 0 to 10.
8. Define a function overlapping() that takes two lists and returns True if they have at least one member in common, False otherwise.
9. Write a function find_longest_word() that takes a list of words and returns the length of the longest word and that word itself.
10. . Write a function filter_long_words() that takes a list of words and an integer len and returns the list of words that are longer than len.
11. Write a program to Interchange First and Last Element of a List.
12. Write a python program to print the sum of tuple elements.
13. Replace single element 'b' in given list ['a', 'b', 'c', 'd', 'e'] with [1, 2, 3]
14. write a program to find index of element 'e' in given vowels list ['a', 'e', 'i', 'o', 'i', 'u']
15. Accept the full name from the user(Name Middlename Surname) in lowercase and Print it in title case. NOTE:(Using in-built function)

16. Use inheritance

Write a class Course with name,fees.

Provide following functionalities

- A. initializer
- B. Accept data
- C. Print Data .

Write a Class Computer with subjectList.

Provide following functionalities

- A. initializer
- B. Accept data
- C. Print Data .

Write a Class Electronics with subjectList.

Provide following functionalities

- A. initializer
- B. Accept data
- C. Print Data .

Write menu driven program to test above functionalities.

OutPut

'1st year Comp' 90000 ['sub1','sub2'....]

17. Write a program that contains a function that has one parameter, n , representing an integer greater than 0. The function should return $n!$ (n factorial). Then write a main function that calls this function with the values 1 through 20, one at a time, printing the returned results. This is what your output should look like:

126

24

120

720

5040

40320

362880

36288002.

18. Write a version of a palindrome recognizer that also accepts phrase palindromes such as "Go hang a salami I'm a lasagna hog.", "Was it a rat I saw?", "Step on no pets", "Sit on a potato pan, Otis", "Lisa Bonet ate no basil", "Satan, oscillate my metallic sonatas", "I roamed under it as a tired nude Maori", "Rise to vote sir", or the exclamation "Dammit, I'm mad!". Note that punctuation, capitalization, and spacing are usually ignored.

19. Given a dictionary of students and their favorite colors:

```
people={'Arham':'Blue','Lisa':'Yellow','Vinod':'Purple','Jenny':'Pink'}
```

A. Find out how many students are in the list

B. Change Lisa's favorite color

C. Remove 'Jenny' and her favorite color

D. Sort and print students and their favorite colors alphabetically by name

20. Create a class 'Student' with roll no, studentName, course ,dictionary of marks(subjectName ->marks [5]).

Provide following functionalities

A. initializer

B. override `__str__` method

C. accept student data

D. Print student data for given id.

E. Print Student who has failed in any subject.

Write menu driven program to test above functionalities.(accept records of 5 students and store those in list)

21. Write a class Date with day,month,year.

Provide following functionalities

A. initializer

B. Accept date(1/09/2022)

C. validate date (If not then throw exception not InvalidDate).

D. Print Date in format of '1 Sep 2022'.

22. Write a Class Painting with the method calculatePaintingCost.

Write a Class FlatPainting with noOfRooms which is inherited from Class Painting.

Override calculatePaintingCost.(Assume per room painting cost is 10000)

Write a Class BulidingPainting with noOfFlats which is inherited from Class Painting.

Override calculatePaintingCost.(Assume per room painting cost is 25000)

Write main() method to accpet painting assignment with option 1.Flat 2.Building and then calculate the cost according to the type.